Negative Atomic Ions In Extended Kohn-Sham Theory

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Abstract

Within the framework of the ***Extended Kohn-Sham (EKS) model***, we study the ability to ***bind additional electrons*** to ***isolated atoms***. This allows us to investigate the ***ionization conjecture*** numerically and to analyse the behavior of approximate exchange-correlation functionals under fractional electronic charge. We developed a ***Julia code*** that employs a carefully chosen ***finite element basis***, achieving ***arbitrarily high accuracy*** both in the discretization (basis set) and in the finite-precision arithmetic, to represent the electronic states.

Motivation

Why study ionization of atoms in DFT ?

- Ionization conjecture :
  - Statement: "A neutral atom can bind at most one or two electrons"
  - No theoretical proof of the existence of negatives ions for some models.
  - Only upper bound of the maximal number of electrons $N_{max}(z)$ the atom of atomic number z can bind : $z \leq N_{max}(z) \leq C_1 z + C_2$ ([1] and references therein).
- Study of the behaviour of approximate density functional in presence of fractional electronic charge [2].

Model

We consider an atom of nuclear charge $z \in \mathbb{N}$ with $N \in \mathbb{R}^+$ electrons.

Extended-Kohn-Sham Model [3]

$$\underset{\gamma \in \mathcal{P}_N}{\text{minimize}}\, E_{z,N}(\gamma)$$

where :

- the total energy is :

$$E_{z,N}(\gamma) := \underbrace{\mathrm{Tr}\left(-\frac{1}{2}\Delta\gamma\right)}_{\text{Kinetic}} \underbrace{- z\int \frac{\rho_\gamma}{|\cdot|}}_{\text{Nucleous-electrons}} + \underbrace{\frac{1}{2}\int \frac{\rho_\gamma(x)\rho_\gamma(y)}{|x-y|}\mathrm{d}x\mathrm{d}y}_{\text{Hartree}} + \underbrace{E_{xc}(\gamma)}_{\text{Exchange-correlation}},$$

- $\mathcal{P}_N$ is the set of one-body density matrix γ such that one can write

$$\gamma^{\sigma\sigma} = \sum_i n_i^\sigma |\phi_i^\sigma\rangle\langle\phi_i^\sigma|$$

with $\sum_{i,\sigma} n_i^\sigma = N$, $0 \leq n_i^\sigma \leq 1$, $\langle\phi_i^\sigma|\phi_j^\sigma\rangle = \delta_{ij}$, $\nabla\phi_i^\sigma \in L^2$, and under the spin-collinear assumption $\gamma = \begin{pmatrix} \gamma^{\uparrow\uparrow} & 0 \\ 0 & \gamma^{\downarrow\downarrow} \end{pmatrix}$.

Note : We assume no spherical symmetry breaking (which is valid for E_{xc}^{exact} and $E_{xc}^{\text{reduced Hartree-Fock}}$) $\Rightarrow$ one can choose $\boxed{\phi_i^\sigma = \phi_{nlm}^\sigma = \frac{f_{nl}^\sigma(r)}{r} Y_{l,m}(\theta,\varphi)}$.

Discretization

One only needs to discretize the radial part of a finite set of orbitals:

$$f_{nl}^\sigma(r) = \sum_{j=1}^{N_{basis}} U_j^{nl\sigma} Q_j(r)$$

for $0 \leq l \leq l_h$ and $l < n \leq n_h + l$, where

- $(Q_i)_{1\leq i\leq N_{basis}}$ is the radial basis,
- $(U_j^{nl\sigma})$ are unknown and satisfies the constraints

$$\left(U^{nl\sigma}\right)^T S U^{n'l'\sigma} = \delta_{nn'}\delta_{ll'}$$

with S the overlap matrix.

Discretization Parameters

- l_h : Max angular momentum,
- n_h : Max principal quantum number,
- R_{max} : Radial cutoff,
- N_{basis} : Cardinality of the radial basis,
- N_{mesh} : Size of the radial mesh,
- s : Mesh parameter,
- D : Max degree of polynomials used.

Radial Basis : Finite Element method

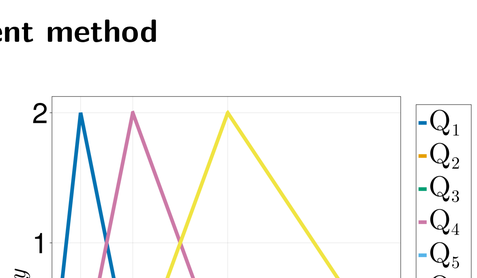
Exponential Mesh :

$$r_i = (1 + R_{max})^{\left(\frac{i}{N_{mesh}-1}\right)^s} - 1$$

with s a parameter.

Generators (defined on $[-1, 1]$)

- Hat functions (piecewise linear)
- Integrated Legendre polynomials (order 2 to D)
  - $P_2(x) = \frac{1}{2}(x^2 - 1)$,
  - $P_3(x) = \frac{1}{2}x(x^2 - 1)$,
  - $P_{n+2}(x) = \frac{2n+1}{n+1}\int_{-1}^{x} xP'_{n+1}(x)\mathrm{d}x - \frac{n}{n+1}P_n$.

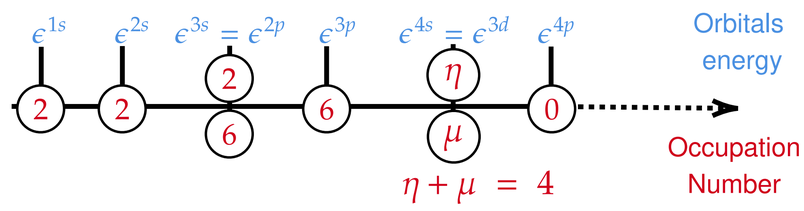


Algorithm

To determine γ^{opt}, one must solve a non-linear generalized eigenvalue problem. We employ the ***optimal damping algorithm (ODA)*** to iteratively converge to the solution.

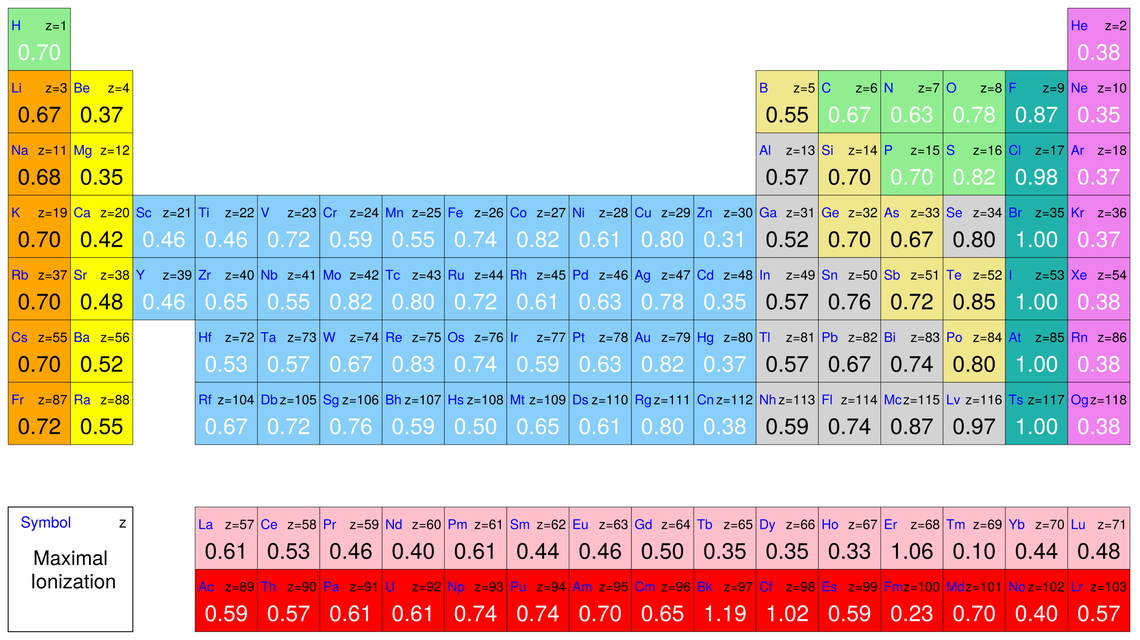
While $||\rho_{\gamma^k} - \rho_{\gamma^{k-1}}|| > tol$ **: do step k :**

1. Using γ^k, compute the Exchange-correlation matrix by quadrature and the Hartree matrix by solving the electrostatic problem and assemble the mean field Hamiltonian $H_{z,N}(\rho_{\gamma^k})$
2. For each l and σ, solve the generalized eigenvalue problem
$$H_{z,N}^{l,\sigma}(\rho_{\gamma^k})U^{l,\sigma} = \epsilon^{l,\sigma} S U^{l,\sigma}$$
3. Aufbau principle to fill n_{nl}^σ : Example with 22 electrons for LDA. When there is a degeneracy, the repartition of electrons is such that it minimizes the energy.

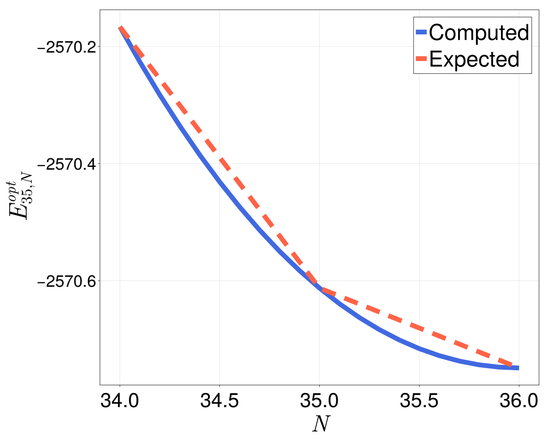


4. Compute γ^* using $(U^{nl\sigma})$ and $(n^{nl\sigma})$ of the current state k.
5. Compute γ^{k+1} by relaxation using the ODA principle :
$$\gamma^{k+1} = t^k\gamma^* + (1 - t^k)\gamma^k$$
where $t^k \in [0, 1]$ is the minimizer of $t \in [0, 1] \mapsto E_{z,N}^{Dis}(t\gamma^* + (1 - t)\gamma^k)$.

Maximal Ionization for LSDA



- We compute numerically $N_{max}(z)$ for the PW exchange correlation using $D = 10$, $s = 1.3$, and $N_{mesh} = 20$. The value of R_{max} is adapted to z by examining the density of the neutral system $N = z$.
- We can analyse the behaviour of E_{xc} in the presence of fractional charge (see [4]).



Further Work

In the future, we plan to:

- Implement other exchange–correlation functionals, such as GGA, meta-GGA, and hybrids,
- Implement a quadratic convergence algorithm to accelerate computations,
- Allow spherical symmetry and spin-collinearity breaking.

References

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